

no ground truth is used; $\log_2$ probes, not a scan

take S of the batch's own queries

answer them at a generous $\alpha_{\max}$
keep that as the reference

bisect the grid: answer the same S at the
midpoint, count top- k slots that differ

within tolerance:
search below it

not within:
search above it

run the rest of the batch at α^*

